

FINPILOT

AI-Powered SME Lending Assistant

**Smarter Credit Decisions
for a Stronger Pakistan**

Analyze Assess Comply Support Empower

Agenda

From data to opportunity

- 01 Problem Statement
- 02 Solution Overview
- 03 System Architecture
- 04 AI Agents
- 05 External Data Sources
- 06 Human-in-the-Loop
- 07 Demo Application
- 08 Technology Stack
- 09 Results & Impact
- 10 Future Enhancements
- 11 Conclusion

The Problem

SMEs in Pakistan face significant challenges in accessing finance.

Limited Access to Credit

Manual & Time-Consuming Assessment

High Risk of Fraud and Misreporting

Complex Regulatory Requirements

Result: Many viable businesses remain unfunded, limiting economic growth.

Our Solution

FINPILOT leverages AI agents, external financial data, and human expertise to provide intelligent, transparent and compliant lending support.

Automated Document Analysis

AI-Powered Financial Assessment

Fraud & Anomaly Detection

Regulatory Compliance (SBP Policies)

A single intelligent platform for faster, fairer and more informed credit decisions.

System Architecture

LangGraph orchestrates the multi-agent pipeline



AI Agents Overview

Specialized agents work together to provide comprehensive analysis.

Document Intelligence Agent

Extracts and analyzes financial document information.

Financial Analysis Agent

Evaluates financial health, ratios and KPIs.

Fraud & Anomaly Agent

Detects suspicious patterns and potential fraud.

Credit Risk Agent

Assesses creditworthiness using multi-factor analysis.

SBP Policy Compliance Agent

Checks compliance with relevant SBP requirements.

External Tools Agent

Fetches external data from PakDataHub and World Bank.

External Data Sources

FINPILOT integrates trusted external APIs for enriched analysis.

PakDataHub

Company profiles, financial data and sector information (Pakistan).

World Bank API

Macroeconomic indicators, country risk data and global benchmarks.

Additional Sources

Industry benchmarks, credit rating data and open datasets.

✓ Real-time, reliable data for better decision making.

Human-in-the-Loop

Domain experts can review and override recommendations when needed.

AI Recommendation

Structured analysis and recommendation.

Human Review

Review analysis • Add comments • Request clarification • Override decision

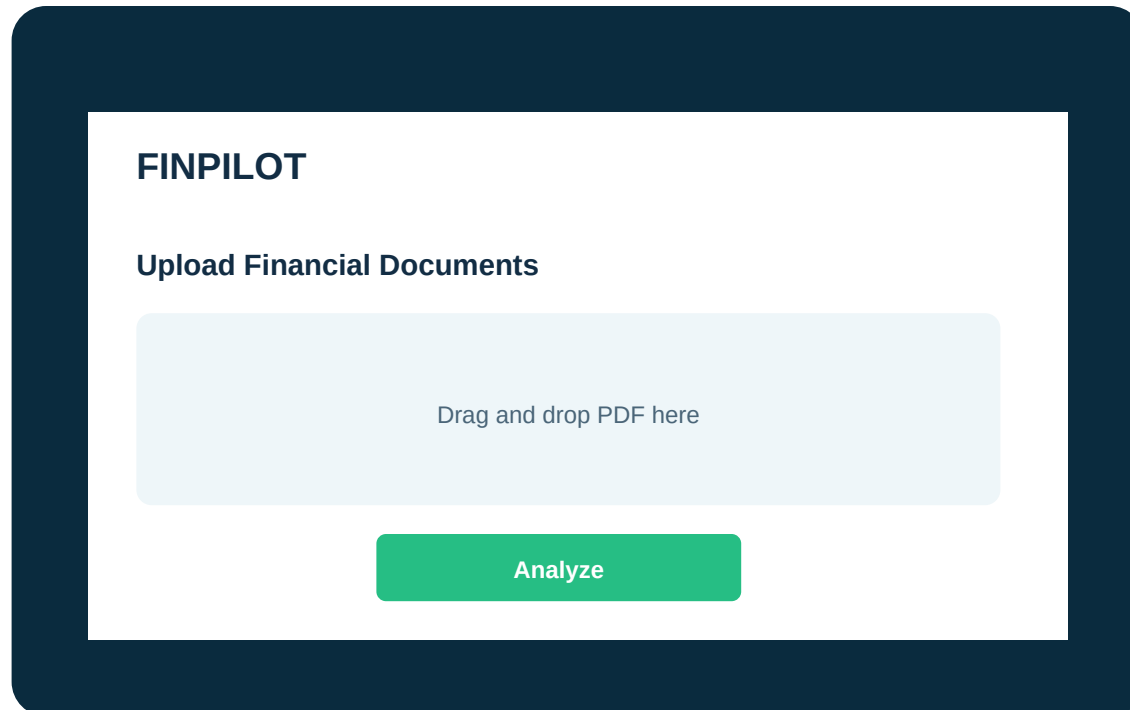
Final Decision

Human-controlled lending outcome.

Combines the speed of AI with the judgment of human experts.

Demo Application

User-friendly interface built with Gradio and FastAPI.

The screenshot shows a web application titled "FINPILOT" with a subtitle "Upload Financial Documents". Below the subtitle is a large light blue rectangular area with the text "Drag and drop PDF here". At the bottom of this area is a green button labeled "Analyze". The entire interface is framed by a dark blue border.

FINPILOT

Upload Financial Documents

Drag and drop PDF here

Analyze

- ✓ Simple and intuitive UI
- ✓ Upload financial documents
- ✓ View agent analysis
- ✓ Human review
- ✓ Final recommendation

Technology Stack

Modern open-source technologies for a scalable solution.

Python

LangChain

LangGraph

FastAPI

Gradio

Pandas

Docker

GitHub

External APIs

Results & Impact

FINPILOT enables faster, more transparent SME lending.

Faster Assessment

From days to minutes

Improved Accuracy

Data-driven analysis

Reduced Fraud Risk

Anomaly detection

Greater Financial Inclusion

Supports viable SMEs

Empowering SMEs today for a stronger, more inclusive tomorrow.

Future Enhancements

Expanding FINPILOT's reach and capabilities.

- ✓ Support for Urdu document processing
- ✓ Integration with additional financial data sources
- ✓ Advanced machine learning models
- ✓ Deployment on cloud (AWS / Azure)
- ✓ Mobile application
- ✓ Multi-language support
- ✓ Continuous learning from feedback

**Bigger
Possibilities
Ahead**

Conclusion

How AI, data and human expertise can work together for SME lending.

Intelligent

AI-assisted analysis

Transparent

Structured evidence

Inclusive

Support for viable SMEs

**FINPILOT is not just a tool — it is a step towards
a stronger, more prosperous Pakistan.**

Thank You

Together for a Brighter
Financial Future



FINPILOT • AI-Powered SME Lending Assistant

[GitHub](#) • [Demo](#) • [Documentation](#)

FINPILOT

AI-Powered SME Lending Assistant

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